

INTELLIGENT AGENTS

Viet-Tuan Le
tuan.lv@ou.edu.vn
vt-le.github.io

1

AGENTS AND ENVIRONMENTS

Definitions

Viet-Tuan Le

2

2

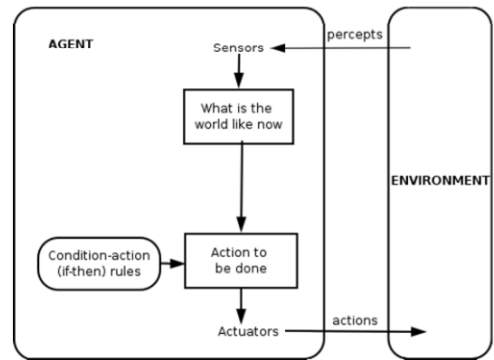
Definitions: agents and environments

- An **agent** is anything that can be viewed as **perceiving** its environment through sensors and acting upon that environment through actuators.

- The **agent function** maps from percept histories to actions:

$$f: \mathcal{P}^* \rightarrow \mathcal{A}$$

- The **agent program** runs on the physical architecture to produce the agent function f



https://en.wikipedia.org/wiki/Intelligent_agent

3

3

Definitions: agents and environments

- The agent program runs on the physical architecture to produce the agent function

$$\text{agent} = \text{architecture} + \text{program}$$

4

Definitions: agents and environments

- Example 1: **human** agent
 - Sensors: eyes, ears, ...
 - Actuators: hands, legs, mouth, ...
- Example 2: **robotic** agent (e.g., Aishimo)
 - Sensors: camera, infrared range finders
 - Actuators: various motors
- Example 3: **software** agent
 - Sensors: keypad, file uploader, network packet receiver...
 - Actuators: screen, file writer, network packet sender...

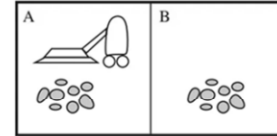
Definitions: agents and environments

- The agent's percept sequence is the complete history of percepts received by the agent
- In general, an agent acts based on its percept sequence
 - But, of course, it cannot act on something it did not perceive
- The “**rationality**” of an agent's behavior depends a lot on its environment, and on its percept sequence
 - Some environments are more difficult than others
 - Percepts in a percept sequence might be “contradictory”

Example: Vacuum-cleaner world

- Percepts:

- Location (A or B)
- Contents (clean or dirty)



- Actions:

- Left
- Right
- Suck
- NoOp

Percept sequence	Action
[A, clean]	Right
[A, dirty]	Suck
[B, clean]	Left
[B, dirty]	Suck
[A, clean] [A, clean]	Right
[B, clean] [A, dirty]	Suck

Viet-Tuan Le

7

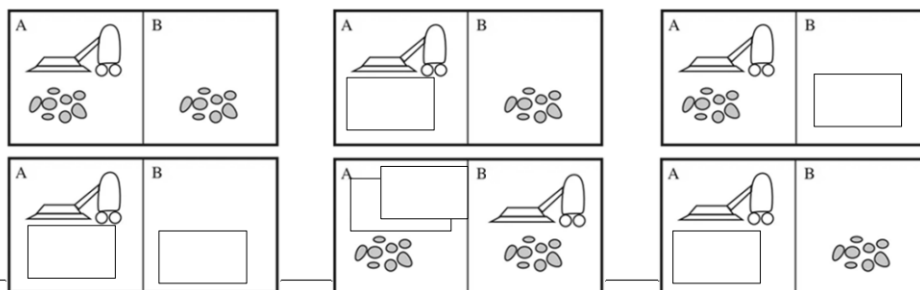
7

Notion of state

- More generally, many things are often not directly completely observable but can be “guessed” from the percept history

- It is the case of the state:

- Set of propositions / diagram that completely describes the environment
- Examples of possible states for the vacuum-cleaner world:



Viet-Tuan Le

8

8

Vacuum-cleaner world

- A simple agent function is:

```
If the current square is dirty, then suck;  
otherwise, move to the other square.
```

- ... or as pseudo-code:

```
function Reflex-Vacuum-Agent( [location,status] ) returns an action  
  if status = Dirty then return Suck  
  else if location = A then return Right  
  else if location = B then return Left
```

- How do we know if this is a good agent function?
- What is the best func?
- Is there one?
- Who decides this?

Rationality

- Fixed performance measure evaluates the environment sequence
 - one point per square cleaned up in time T?
 - one point per clean square per time step, minus one per move?
 - penalize for > k dirty squares?
- A rational agent chooses any action that
 - maximizes the expected value of the performance measure
 - given the percept sequence to date
- Rational $\neq$ omniscient
 - percepts may not supply all relevant information
- Rational $\neq$ clairvoyant
 - action outcomes may not be as expected
- Hence, rational $\neq$ successful
- Rational $\Rightarrow$ exploration, learning, autonomy

PEAS

- To design a rational agent, we must specify the **task environment**, which consists of the following four things:
 - **Performance measure??**
 - **Environment??**
 - **Actuators??**
 - **Sensors??**
- Examples of agents:
 - Automated taxi,
 - Internet shopping agent,
 - Boardgames

11

Automated taxi

- The task environment for an automated taxi:
 - **Performance measure??**
 - **Environment??**
 - **Actuators??**
 - **Sensors??**



12

Vacuum cleaner

- Performance measure??
- Environment??
- Actuators??
- Sensors??



Viet-Tuan Le

13

13

Internet shopping agent

- The task environment for an internet shopping agent:
 - Performance measure??
 - Environment??
 - Actuators??
 - Sensors??

Viet-Tuan Le

14

14

Question-answering system

- The task environment for a question-answering system:
 - Performance measure??
 - Environment??
 - Actuators??
 - Sensors??



Viet-Tuan Le

15

15

Question-answering system: Jeopardy!

- The task environment for a question-answering system:
 - Performance measure?? \$ \$ \$
 - Environment?? Wikipedia, Wolfram alpha, ontologies, encyclopedia, . . .
 - Actuators?? Spoken/written language, answer button
 - Sensors?? Written/spoken input
- Why did IBM choose Jeopardy as its goal for a QA system?

Because of the performance measure!



Viet-Tuan Le

16

16

Environment types

- Fully observable vs partially observable
 - Fully observable gives access to complete state of the environment
 - Complete state means aspects relevant to action choice
 - Example: global vs local dirt sensor
- Deterministic vs Stochastic
 - Deterministic – next state completely determined by current state and action
 - Uncertainty may arise because of defective actions or partially observable state (i.e., agent might not see everything that affects the outcome of an action).

Environment types

- Episodic vs Sequential
 - Episodic the agent's experience divided into atomic episodes
 - Next episode not dependent on actions taken in previous episode. E.g., assembly line
 - Sequential – current action may affect future actions. E.g., playing chess, taxi
 - Short-term actions have long-term effects
 - Must think ahead in choosing an action

Environment types

- Static vs Dynamic

- Does environment change while agent is deliberating?
- Static - crossword puzzle
- Dynamic - taxi driver

- Discrete vs Continuous

- Can refer to
 - The state of the environment (chess has finite number of discrete states)
 - The way time is handled (taxi driving continuous – speed and location of taxi sweep through range of continuous values)
 - Percepts and actions (taxi driving continuous – steering angles)

Environment types

- Single Agent vs Multi-agent

- Single Agent – crossword puzzle
- Multi-agent – chess, taxi driving? (are other drivers best described as maximizing a performance element?)
- Multi-agent means other agents may be competitive or cooperative and may require communication
- Multi-agent may need communication

Environment types

- The environment type largely determines the agent design
 - The real world is (of course) partially observable, stochastic, sequential, dynamic, continuous, multi-agent

	Solitaire	Backgammon	Web shopping	Taxi
Observable??				
Deterministic??				
Episodic??				
Static??				
Discrete??				
Single-agent??				

Viet-Tuan Le

21

21

Agent types

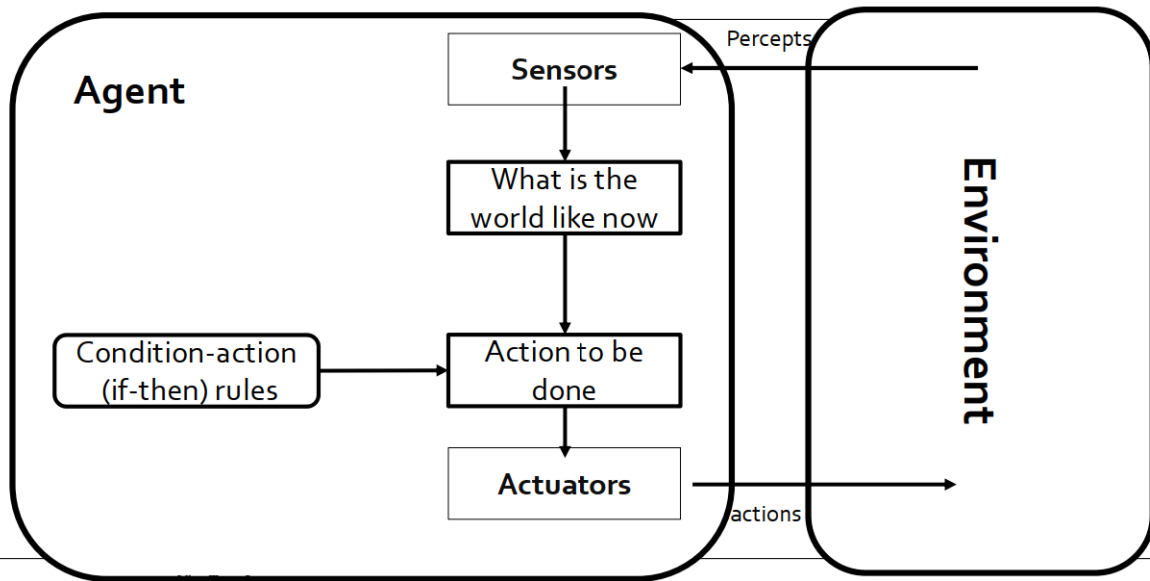
- Four basic types in order of increasing generality:
 - Simple reflex agents
 - Reflex agents with state
 - Goal-based agents
 - Utility-based agents
- All these can be turned into learning agents

Viet-Tuan Le

22

22

Simple reflex agents



23

23

Example

```
function Reflex-Vacuum-Agent( [location,status]) returns an action
  if status = Dirty then return Suck
  else if location = A then return Right
  else if location = B then return Left
```

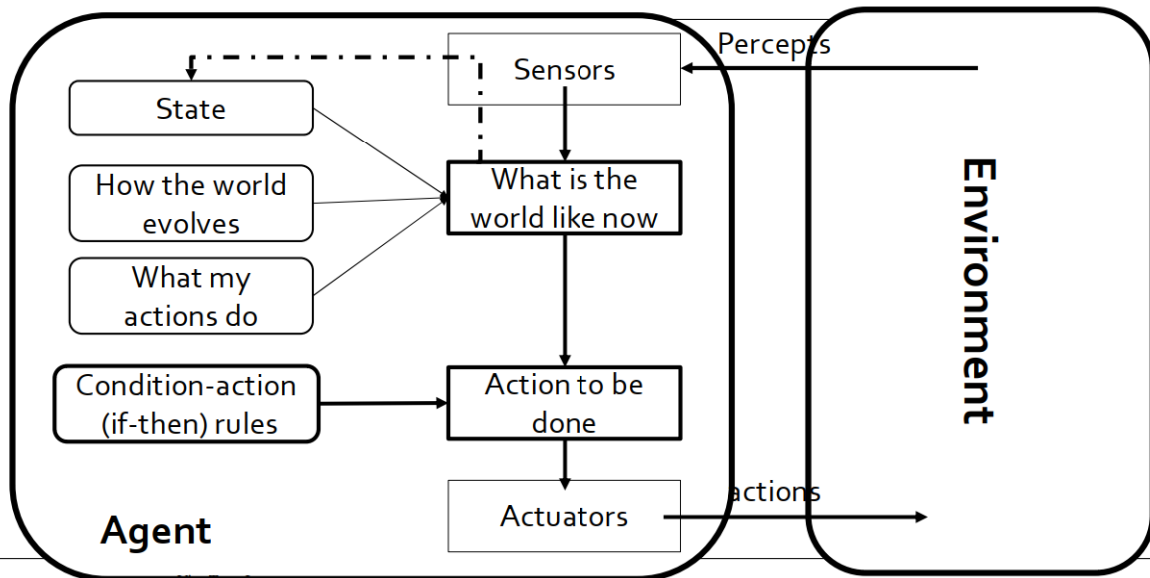
```
def reflex_vacuum_agent(sensors):
    if sensors['status'] == 'Dirty':
        return 'Suck'
    elif sensors['location'] == 'A':
        return 'Right'
    elif sensors['location'] == 'B':
        return 'Left'
```

Viet-Tuan Le

24

24

Reflex agents with state



25

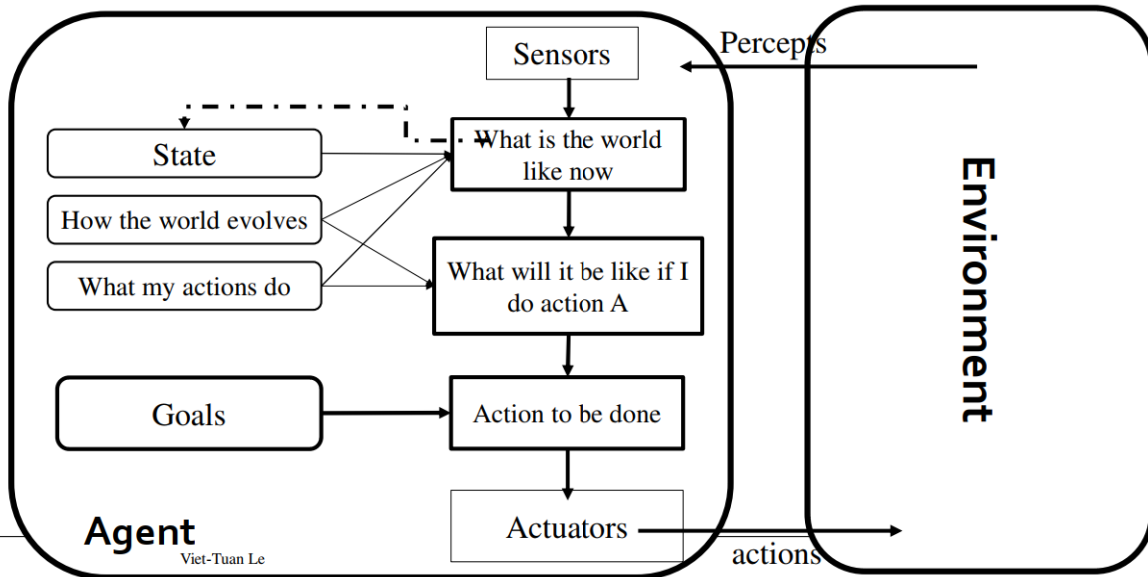
Example

```
function Reflex-Vacuum-Agent( [location,status]) returns an action
Static: last_A, last_B, numbers, initially ∞
    if status = Dirty then ...

initial_state = {'last-A': maxint, 'last-B': maxint}
def reflex_vacuum_agent_state(sensors, state=initial_state):
    if sensors['status'] == 'Dirty':
        if sensors['location'] == 'A':
            state['last-A'] = 0
        else:
            state['last-B'] = 0
        return 'Suck'
    elif sensors['location'] == 'A' and state['last-B'] > 3:
        return 'Right'
    elif sensors['location'] == 'B' and state['last-A'] > 3:
        return 'Left'
    else:
        return 'NoOp'
```

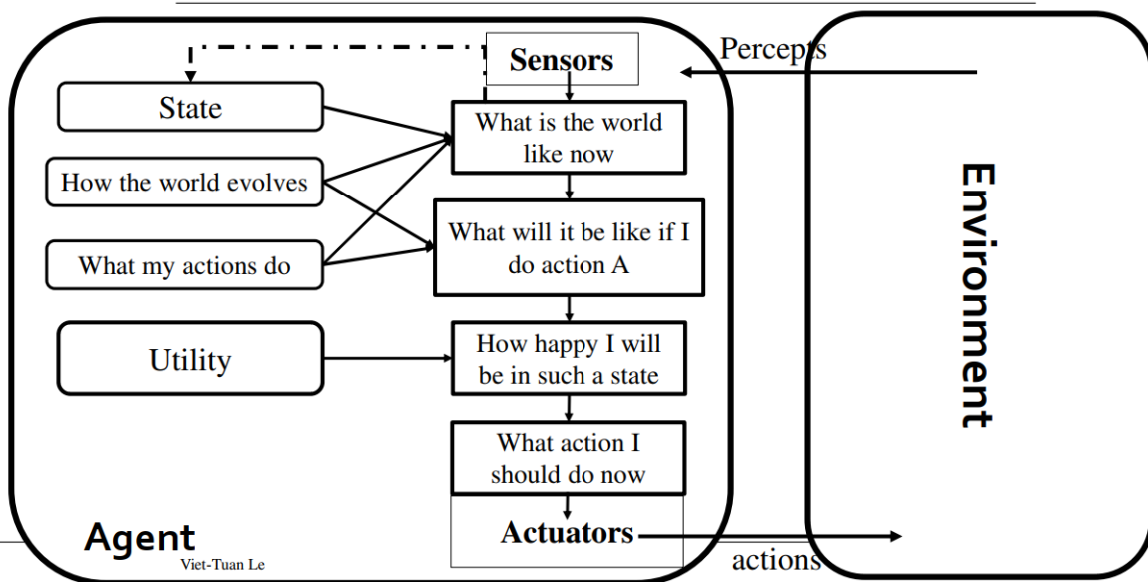
26

Goal-based agents



27

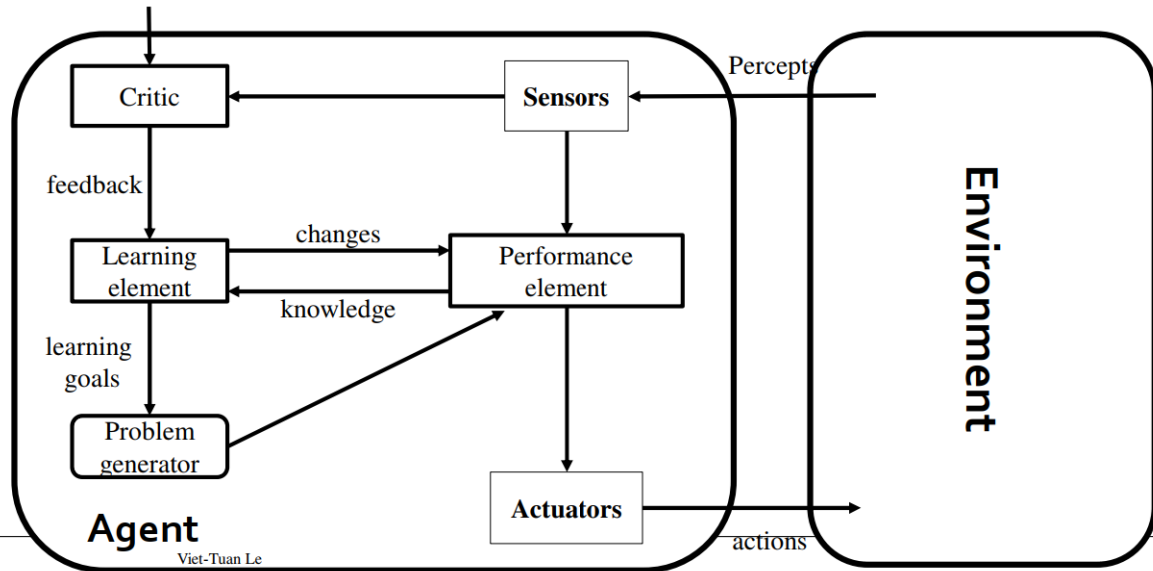
Utility-based agents



28

Learning agents

Performance standard



Viet-Tuan Le

29

29

Summary

- Agents interact with environments through actuators and sensors
- The agent function describes what the agent does in all circumstances
- The performance measure evaluates the environment sequence
- A perfectly rational agent maximizes expected performance
- Agent programs implement (some) agent functions
- PEAS descriptions define task environments
- Environments are categorized along several dimensions:
observable? deterministic? episodic? static? discrete? single-agent?
- Several basic agent architectures exist:
reflex, reflex with state, goal-based, utility-based

Viet-Tuan Le

30

30

Q&A

Thank you!

Viet-Tuan Le

31